

APEX. CARWASH SYSTEM

Fullstack Architecture & Technical Specifications (Project 5)

Backend: Express + Prisma + PostgreSQL  
Payment: Midtrans Snap (QRIS/VA) + Kasir • Realtime: Socket.IO

1. Arsitektur Dua Sisi & User Journey

Sisi Customer (Portal Mandiri)

- **Autentikasi Email:** Registrasi/Login instan tanpa password.
- **Vehicle Self-Service:** Tambah, lihat, dan kelola mobil milik sendiri.
- **Booking Mandiri:** Booking cuci dengan proteksi anti-double booking.
- **Pembayaran Online Midtrans:** Bayar via Snap QRIS, Virtual Account (BCA/Mandiri/BRI/BNI), atau Bayar di Kasir.
- **Live FIFO Tracking:** Posisi antrean ( `queuePosition` ), mobil di depan ( `ahead` ), dan estimasi waktu tunggu realtime.
- **Faktur Digital:** Struk resmi lunas + tombol cetak & unduh PDF.

Sisi Admin / Kasir (Protected)

- **Autentikasi Admin:** Login via email ( `admin@carwash.com` ) & password.
- **Order Management:** Monitoring antrean, update status ( `IN_PROGRESS` , `COMPLETED` ), dan assign staf cuci ( `staffId` ).
- **Kasir & Billing:** Proses pembayaran kasir (CASH/QRIS/TRANSFER) & cetak invoice.
- **Master Data:** Layanan cuci, 8 Staf (7 aktif + 1 off), Customer, & Kendaraan.
- **Dashboard Metrik:** Total Pelanggan, Omzet Lunas, Transaksi, & Antrean Aktif.

2. Status Order, Status Pembayaran & Integrasi Midtrans

WAITING

Order masuk antrean FIFO. `staffId`: `null`.

IN\_PROGRESS

Mobil masuk cuci. Admin menugaskan `staffId`.

COMPLETED

Cuci selesai, `completedAt` terisi, siap diambil.

CANCELLED

Dibatalkan oleh Customer (saat WAITING) / Admin.

Payment Gateway Midtrans Snap & Kasir

- **Midtrans Snap Token:** `POST /api/payments/midtrans-token` mengembalikan `snapToken` untuk pop-up checkout di frontend (QRIS, VA BCA/Mandiri/BRI/BNI, GoPay).
- **Webhook Notification:** `POST /api/payments/midtrans-webhook` menangani callback otomatis dari Midtrans untuk mengubah `paymentStatus`: `"PAID"` dan generate invoice.
- **Kasir Manual:** Pembayaran tunai/manual di kasir ( `POST /api/orders/:id/pay` ).

Kalkulasi Antrean FIFO

- `queuePosition`: Urutan antrean di antara order WAITING.
- `estimatedMinutes`: Durasi mobil IN\_PROGRESS + mobil WAITING di depannya.

3. Identitas Visual & Design System

Deep Obsidian

#07080c (Dark Bg)

Neon Purple

#a855f7 (Accent/Brand)

Emerald Lunas

#10b981 (PAID / Ready)

Warm Amber

#f59e0b (In Progress)

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Backend REST API Endpoints, Payment Gateway & Realtime Specs

Port: 3000 • Base Path: /api  
Source of Truth: Database PostgreSQL

4. Pemetaan Lengkap Endpoint REST API Backend

| Endpoint                        | Method | Akses       | Request Body / Params                                    | Fungsi & Output                                   |
|---------------------------------|--------|-------------|----------------------------------------------------------|---------------------------------------------------|
| /api/auth/login                 | POST   | Public      | { email, password }                                      | Login Admin \$→\$ Token JWT (Role ADMIN)          |
| /api/auth/customer/login        | POST   | Public      | { email }                                                | Login Customer \$→\$ Token JWT (Role CUSTOMER)    |
| /api/auth/customer/register     | POST   | Public      | { name, email, phone }                                   | Registrasi Akun Customer Baru                     |
| /api/customer/vehicles          | GET    | Customer    | -                                                        | Daftar kendaraan milik customer                   |
| /api/customer/vehicles          | POST   | Customer    | { plateNumber, brand, model, color, year? }              | Tambah kendaraan baru milik customer              |
| /api/customer/vehicles/:id      | DELETE | Customer    | -                                                        | Hapus kendaraan (dicegah jika sedang antre)       |
| /api/customer/orders            | POST   | Customer    | { vehicleId, services: [{serviceId, quantity}], notes? } | Buat order booking cuci baru (Status: WAITING)    |
| /api/customer/orders/active     | GET    | Customer    | -                                                        | Ambil antrean aktif & queueInfo (FIFO)            |
| /api/customer/orders/history    | GET    | Customer    | -                                                        | Riwayat order selesai (COMPLETED / CANCELLED)     |
| /api/customer/orders/:id/cancel | PATCH  | Customer    | -                                                        | Batalan order mandiri (hanya jika masih WAITING)  |
| /api/payments/midtrans-token    | POST   | Customer    | { orderId }                                              | Request Snap Token Midtrans (QRIS/VA)             |
| /api/payments/midtrans-webhook  | POST   | Midtrans    | Midtrans notification payload                            | Callback status bayar (ubah PAID otomatis)        |
| /api/orders                     | GET    | Admin       | -                                                        | Ambil seluruh data order di sistem                |
| /api/orders/:id                 | PUT    | Admin       | { status?, staffId?, notes? }                            | Update status cuci & assign staf bertugas         |
| /api/orders/:id/pay             | POST   | Admin       | { paymentMethod, paidAmount? }                           | Bayar di kasir, ubah PAID & buat Invoice          |
| /api/orders/:id/invoice         | GET    | Auth        | -                                                        | Ambil data invoice resmi untuk dicetak/PDF        |
| /api/services                   | GET    | Public/Auth | -                                                        | Katalog paket layanan cuci & harga                |
| /api/staff                      | GET    | Admin       | -                                                        | Daftar seluruh staf cuci (8 Staf: 7 aktif, 1 off) |

5. Arsitektur WebSocket Realtime Tracking (Socket.IO)

- Alur Realtime Event & Sinkronisasi:
- Initial State (REST):** Frontend memuat data antrean awal via GET /api/customer/orders/active.
  - Socket Connection:** Client terhubung ke server Socket.IO pada port yang sama (HTTP port 3000).
  - Event Broadcast:** Ketika Admin mengubah status order atau Customer membayar via Midtrans, backend memancarkan event ORDER\_STATUS\_UPDATED dan QUEUE\_UPDATED.
  - Reconnect Resilience:** Saat koneksi internet terputus lalu kembali, frontend melakukan re-fetch REST API untuk sinkronisasi mutlak.

6. Roadmap Eksekusi Terpadu (9 Step)

- Step 1: REST API + Booking + FIFO (Done ✓)**  
Autentikasi dual role, vehicle CRUD, booking, dan FIFO queue.
- Step 2: Setup Socket.IO Server (Next 🚧)**  
Integrasi Socket.IO pada Express HTTP server port 3000.
- Step 3: Test Connection Sederhana**  
Uji coba koneksi client-server websocket handshake.
- Step 4: Implement Order Room**  
Isolasi event per customer / order room.
- Step 5: Event ORDER\_STATUS\_UPDATED**  
Broadcast status order saat diproses oleh Admin.

- Step 6: Event QUEUE\_UPDATED**  
Kalkulasi ulang dan update antrean customer realtime.
- Step 7: Payment Gateway Midtrans Integration**  
Snap Token endpoint, checkout QRIS/VA, & webhook handler.
- Step 8: Frontend Realtime Tracking & Payment**  
UI live stepper, Midtrans Snap dialog, dan invoice PDF.
- Step 9: Reconnect Resilience & Auth Guard**  
Sinkronisasi otomatis saat reconnect & validasi JWT.